Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/26/2008
PI Name:	Hangarter, Roger P Ph.D.		
Project Title:	Gravity response mechanisms of lateral organs and the control of plant architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	47405-7107	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:	When this project was terminated prematurely in response to the President's vision to send humans to Mars, the postdoc and student left my lab to seek more stable funding. Due to the short time remaining on the project after the termination notice, it was not possible to hire anyone to continue the research.		
COI Name (Institution):	Mullen, Jack (Indiana University)		
Grant/Contract No.:	NNA04CC55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We demonstrated that the orientation of lateral organs in plants (roots, branches and leaves) is determined by what is called the gravitropic set-point angle (GSA). That is, the specific orientation of lateral organs is genetically determined but controlled by the gravitropic responses of the lateral organs. The goal of this project was to use a combination of genetic, molecular, and physiological approaches to better understand the role of gravity in guiding whole plant development. Specifically, we used Arabidopsis as a model system for identifying genes required for controlling the GSA of later roots, rosette leaves and inflorescence branches. We have isolated a number of mutants that display altered GSA responses in lateral organs. The mutants have been designated mgsa mutants for modified gravitropic set-point angles. The molecular and physiological analyses of the mgsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. To discover and characterize genes that are required for controlling the orientaion of lateral organs during plant development and to understand the mechanisms by which plants sense and respond to gravity. The molecular and physiological analyses of the gsa mutants will provide new insight into the basis of GSA regulation and, thus, a better understanding of the physiological and molecular mechanisms by which plants use gravity to control their architecture. Since the bulk of a plant consists of lateral organs, our studies will contribute to a better understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for growth on future long-term space missions or for agriculture.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since the bulk of a plant consists of lateral organs, our studies will contribute to our understanding of the fundamental developmental biology of plant form and may eventually lead to the development of improved plant varieties for high-density agricultural. Unfortunately, since the project was terminated prematurely by NASA, the impact of this work will not be realized until another funding agency supports the project.
Task Progress:	We identified several genes that, when mutated, affect the orientation of lateral organs in Arabidopsis. A few of the genes specifically affect lateral roots and others affect the orientation of leaves and branches. We were able to map one of the genes, MGSA4 to a small region on chromosome 4. Plants with a mutant version of mgsa4 display vertically oriented leaves and inflorescence branches. Identification of the MGSA4 gene could potentially allow genetic engineering of plants with altered architectures for agronomic and ornamental uses. We characterized the gravitropic behavior of lateral roots and shoots in Arabidopsis and a few other plant species. In all cases, we found that lateral organs orient relative to the gravity vector using the same basic gravitropic response system used buy the primary roots and shoots. However, unlike the primary root and shoot, which typically maintain vertical orientation, lateral organs position themselves more horizontally. Moreover, they change their orientation in a developmentally dependent manner. For example, a young Arabidopsis branch elongates at near 90 degrees from vertical for a few days but then repositions to angles of about 60 degrees from vertical. Our studies showed that these developmental changes in branch orientation result from a fundamental change in the gravitropic set point (GSA) of the organ. Our results provide the most complete analysis of the GSA and demonstrate how plant architecture is regulated by its alteration during normal plant development. We identified several genes that when mutated alter the orientation of lateral organs. We identified four genes that specifically affect orientation of lateral roots and another three that specifically affect leaf and branch orientation. We were able to identify one of the root genes to be an allele of RHD3, which had been previously shown to affect root hair development. We initiated mapping of the other root-specific mutants but the phenotypes are difficult to score and progress was slow. The phenotype of the shoot mutants was more robust so we focused on one of those mutants, mgsa4, for detailed mapping and eventually gene identification. We characterized the phenotype of the mgsa4 mutant in detail and fund that the altered gravitropic response is restricted to the lateral organs of the above ground part of the plant. Specifically, the main shoot displays normal vertical orientation and gravitropic responses during all stages of development. The leaves and lateral branches also display normal gravitropism in response to changes in orientation. That is, if a branch is reoriented, it reorients back to the original position just as robustly as in wild type plants. However, the lateral organs display a basal orientation that is as much as 40 degrees more vertical than wild type. Based on our physiological analyses, we were able to conclude that the mgsa4 gene specifically affects the overall orientation of the lateral organs without affecting the basal gravitropic machinery. Our results represent a novel demonstration of a fundamental aspect of plant gravity responses that has not been described before. Unfortunately, because this project was terminated prematurely, we were not able to identify the gene at the molecular level. Project personnel left the lab and manuscripts that were in early stages of preparation could not be completed.
Bibliography Type:	Description: (Last Updated: 08/11/2025)
Articles in Peer-reviewed Journals	Mullen JL, Hangarter RP. "Genetic analysis of the gravitropic set-point angle in lateral roots of Arabidopsis." Adv Space Res. 2003 May;31(10):2229-36. http://dx.doi.org/10.1016/S0273-1177(03)00249-7 , May-2003
Articles in Peer-reviewed Journals	Mullen JL, Wolverton C, Hangarter RP. "Apical control, gravitropic signaling, and the growth of lateral roots in Arabidopsis." Adv Space Res. 2005;36(7):1211-7. http://dx.doi.org/10.1016/j.asr.2005.03.103 , Sep-2005
Articles in Peer-reviewed Journals	Mullen JL, Weinig C, Hangarter RP. "Shade avoidance and the regulation of leaf inclination in Arabidopsis." Plant Cell Environ. 2006 Jun;29(6):1099-106. PMID: 17080936 , Jun-2006
Articles in Peer-reviewed Journals	Roychoudhry S, Sageman-Furnas K, Wolverton C, Grones P, Tan S, Molnár G, De Angelis M, Goodman HL, Capstaff N, Lloyd JPB, Mullen J, Hangarter R, Friml J, Kepinski S. "Antigravitropic PIN polarization maintains non-vertical growth in lateral roots." Nat Plants. 2023 Sep 4. https://pubmed.ncbi.nlm.nih.gov/37666965 ; PMID: 37666965; PMCID: PMC10505559 , Sep-2023